

TARANJYOT SINGH

Senior Software Engineer | AI Platform Engineer | Python • FastAPI • AWS • GenAI
Mississauga, ON, Canada

taranjyotsingh357@gmail.com | [LinkedIn/Taranjyot Singh](#) | [Github/Taranjyot Singh](#) | [Portfolio/Taranjyot Singh](#)

PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years of experience building cloud-native backend platforms, AI-powered applications, and scalable distributed systems across Financial Services, Travel Technology, and Generative AI domains. Specialized in Python, FastAPI, AWS, Kubernetes, Docker, CI/CD, and modern software architecture, with hands-on experience developing framework-level solutions supporting multiple enterprise products. Proven expertise in LLM evaluation, RLHF, prompt engineering, backend platform development, data engineering, and production-grade AI systems. Passionate about solving complex engineering challenges, enabling developer productivity through platform engineering, and delivering secure, scalable software that drives measurable business impact.

SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, C++, Bash, SQL
- **Backend & Platform Engineering:** FastAPI, Flask, Django, REST APIs, Microservices, Distributed Systems, Pydantic, Celery, Event-Driven Architecture
- **AI / GenAI:** LLMs, RLHF, Prompt Engineering, RAG, Embeddings, Vector Databases, AI Agents, Agentic Workflows, Model Evaluation, NLP, PyTorch, TensorFlow
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, Jenkins, Terraform, GitHub Actions, DevSecOps, CI/CD
- **Data Engineering:** PostgreSQL, MySQL, MongoDB, Elasticsearch, Pandas, NumPy, ETL Pipelines, Data Processing
- **Tools:** Git, GitHub, Bitbucket, Postman, Redis, Grafana, Jira, Confluence, VS Code
- **Software Engineering:** System Design, Platform Engineering, Framework Development, API Design, Software Architecture, Release Management, Agile Development

WORK EXPERIENCE

Generative AI Associate

Aug. 2025 - Present

Innodata | AI Data Engineering & Generative AI Services

Remote, Canada

- Performed LLM evaluation, annotation, and prompt-based assessments to improve accuracy, safety, and alignment of Meta's generative AI and multimodal models.
- Analyzed and graded model outputs for quality, relevance, factual correctness, and policy compliance, contributing to Meta's RLHF and data-training pipelines.
- Executed high-precision data labeling, categorization, and QA workflows using Meta's internal platforms while meeting strict productivity and accuracy targets.
- Ensured data privacy, confidentiality, and security by following all Innodata CNDA requirements and Meta's AI safety policies.
- Utilized Python for analyzing, validating, and processing LLM-generated outputs, including structured data handling, quality assessment, and workflow automation tasks.
- Applied Python-based scripting and analytical techniques to support AI evaluation pipelines, data annotation processes, and QA validation for generative AI models.
- Collaborated with team leads and QA reviewers to maintain consistent task quality, resolve ambiguities, and meet operational SLAs in a fully remote environment.
- **Technical Stack:** Python, NLP, RLHF, Prompt Engineering, LLM Evaluation, Data Annotation, AI Quality Assurance, Meta AI Tooling

Software Engineer - AI Training

Nov. 2024 - Aug 2025

Outlier AI | Generative AI & RLHF Platform

Remote, Canada

- Trained and fine-tuned LLMs using Reinforcement Learning with Human Feedback (RLHF) to enhance model accuracy and responsiveness.
- Evaluated and optimized AI-generated outputs for clarity, relevance, and alignment with user intent using structured rubrics and decision matrices.
- Designed and solved complex programming tasks, creating robust test cases to validate AI reasoning and performance.
- Used Python to develop, validate, and optimize coding solutions, test cases, and algorithmic workflows for AI model

evaluation and RLHF training tasks.

- Leveraged Python for data analysis, prompt evaluation automation, and structured validation of AI-generated outputs to improve model accuracy and consistency.
- Delivered detailed, actionable feedback on code and natural language outputs to improve usability, factual accuracy, and ethical compliance.
- Standardized prompt engineering workflows and conducted QA checks to ensure consistent, high-quality LLM behavior across multiple use cases.
- **Technical Stack:** Python, RLHF, Prompt Engineering, LLM Evaluation, NLP, AI Quality Assurance, Model Validation

Software Development Engineer

Jun. 2022 - May 2024

Anuvu | Aviation Connectivity & In-Flight Entertainment Platform

Remote, Canada

- Developed microservices with Python & FastAPI deployed on AWS EC2 using Docker and Kubernetes for improved deployment consistency and operational efficiency.
- Contributed to a shared engineering framework supporting three enterprise products, enabling reusable services, standardized development practices, and improved engineering productivity.
- Led the initiative to implement CI/CD pipelines with Jenkins, automating the testing and deployment process of applications on AWS instances as a backend and framework developer in a team of 5. Also, integrated DevSecOps tools into the pipeline (e.g., Bandit).
- Developed data transformation modules and optimized high-volume pipelines and relational queries across large datasets using SAS, Pandas, PostgreSQL, and Pydantic.
- Collaborated on the containerization and deployment of 6 scalable microservices using Docker and Kubernetes, contributing to integration testing and deployment automation efforts; achieving 87% code coverage through extensive unit testing.
- Oversaw the official releases for 3 of the live products.
- Built backend automation workflows and reusable framework components that accelerated development and improved consistency across multiple enterprise products.
- **Technical Stack:** Python, FastAPI, AWS (EC2, S3), Kubernetes, Docker, Jenkins, PostgreSQL, Pandas, Pydantic, SAS, CI/CD, DevSecOps, Databricks, SQL Server, Pytest, Git, Bitbucket

Software Development Engineer

Oct. 2021 - May 2022

Solut Pvt. Ltd. | Financial Technology & Business Analytics Solutions

Remote, Canada

- Developed and maintained a Flask-based financial services platform supporting transaction processing, operational analytics, and business-critical workflows across over one million secure records.
- Built and optimized financial reporting and analytics pipelines supporting over one million records, improving data processing efficiency and operational reporting workflows.
- Integrated backend services with MongoDB, Azure Database, and AWS Lambda for serverless processing.
- Partnered directly with business stakeholders to gather requirements, translate financial workflows into technical solutions, and deliver scalable platform enhancements.
- Conducted extensive unit and integration testing using Pytest to ensure application reliability and functionality.
- **Technical Stack:** Python, Flask, Pandas, MySQL, MongoDB, Azure Database, AWS Lambda, Docker, Kubernetes, React, JavaScript, Pytest, Git, CI/CD

Software Development Intern

May 2018 - Dec. 2018

Kyrion Technologies | AI & Computer Vision Solutions

New Delhi, India

- Built an AI-based OpenCV-driven digit recognition system achieving 90% accuracy.
- Deployed a Python utility leveraging the usage of OpenCV to train a version on a dataset of 500 handwritten digits as part of a software program development group.
- Applied TensorFlow and PyTorch models on annotated datasets for real-time prediction use cases.
- **Technical Stack:** Git, OpenCV, matplotlib, Python, Pytest, PyTorch, TensorFlow, Postgres Database, Docker

Software Development Intern

Jun. 2017 - Jan. 2018

Trip String | Travel Technology & Recommendation Platform

New Delhi, India

- Designed and implemented a travel website which utilized user input to automatically rank and present alternative trip packages based on the filters supplied by the user.
- Managed the website's cloud deployment using Linux servers and automated updates using Bash scripting.
- **Technical Stack:** HTML, CSS, JavaScript, React, WordPress, Bash, Linux, Python

EDUCATION

Master of Engineering (Quality Systems Engineering)

Concordia University, Montreal, QC, Canada GPA: 3.3/4.3

Bachelor of Technology (Computer Science and Engineering)

Guru Gobind Singh Indraprastha University, Delhi, India GPA: 7.8/9 - First Division with Distinction

PROJECTS

Resume AI Platform | FastAPI, React, Ollama, RAG, JWT | [TaranjyotS/resume-api](#)

- Built an AI-powered full-stack platform generating ATS-optimized resumes, cover letters, recruiter emails, and interview preparation content using FastAPI, React, Docker, and LLM integrations.

Smart Budget Allocator | FastAPI, React, Financial Planning Engine | [TaranjyotS/smart-budget-allocator](#)

- A full-stack personal finance and Canadian tax planning platform that helps users manage income, expenses, budgeting, financial goals, asset allocation, and optimize TFSA, FHSA, and RRSP contributions in one centralized application.

MLOps CI/CD Platform | MLflow, DVC, FastAPI, GitHub Actions | [TaranjyotS/mlops-ci-cd](#)

- Production-ready MLOps CI/CD framework that automates data validation, model training, experiment tracking, and versioned pipelines using DVC, MLflow, FastAPI, Docker, and GitHub Actions.

Web Security Auditor | .NET 8, Blazor, OWASP, TLS Analysis | [TaranjyotS/web-security-auditor](#)

- Developed a security assessment platform that analyzes websites for security headers, SSL/TLS configuration, certificate validation, and OWASP-related security risks. Implemented automated scanning workflows, security reporting dashboards, and backend analysis services to improve web security visibility and compliance.

PUBLICATIONS

RDBMS Analysis for Big Data using Elasticsearch | [IJRECE, Vol. 7, Issue 2, 2019](#)

- Published research exploring Elasticsearch-based search architectures for Big Data analytics, focusing on indexing, sharding, inverted indexes, and full-text search optimization for scalable data platforms.

CERTIFICATIONS

- **Artificial Intelligence Certification** - (HPE certified) *WAC Institute*
- **Java/Android Development Certification** - *CMC Institute*
- **AWS Cloud Technical Essentials Certification** - *Coursera*
- **Google AI Essentials Certification** - *Coursera*